

NABU COMMAND CENTER

Version 1.0 Community Release White Paper & Publication Update

From a December 2023 forum proposal to a native, hardware-verified NABU information and telemetry console

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STATUS Current status: V1.0 LOCAL FREEZE VERIFIED. The exact accepted .NABU binary has already passed Owner acceptance in MAME and on physical NABU hardware. Derek-owned NABU Command Center material is released under the MIT License, Copyright (c) 2026 Derek Leger; third-party material retains its own terms. This edition removes physical-NABU photographs from public distribution while preserving the verification record.

1. Executive Community Statement

In December 2023 I posted an early idea to the NABU forum under the title “NABU Weather and News Version 1.0.” At that point the concept was deliberately rough: use the NABU as a practical network terminal for current information rather than treating it only as a static preservation platform. The example BASIC listing in that post was exploratory and openly depended on functions and server behavior that had not yet been proven.

The project that emerged is substantially more rigorous. NABU Command Center v1.0 is now a native Z80 application built for the NABU target, paired with a modern companion Gateway that handles contemporary Internet services and reduces their data into small deterministic records for the NABU. The 1980s machine is not being asked to behave like a modern browser; it is doing the work it is good at: input, validation, bounded state, graphics, sound, and a responsive command-center presentation.

Version 1.0 has now completed its local production freeze. The exact accepted client binary was preserved without rebuilding, its source and release evidence were captured in Git, a dedicated release branch and annotated physical-verified tag were created, the release directory was organized, the final engineering white paper and user manual were included unchanged, and Git integrity checks passed. The remaining work before formal community publication is procedural: final licensing/provenance review, creation of the public repository/release, NABU Cloud staging and publication, public-path verification, and the community announcement.

RELEASE IDENTITY The public release will be traceable to one exact native artifact: BUILD_ID
NCC-SPLASH-260826-LV2, 59,501 bytes, SHA-256
5D27DFBB3ED8DCD96AF828CECC7550EF546B07C10509E20A71FECFCF0CC89087.

1.1 Where the project started

Figure 1. Original December 23, 2023 NABU forum proposal. The central idea was a dynamic everyday utility terminal fed by current information.

The original forum post is useful provenance because it shows that the core product idea predates the implementation details. What changed was the engineering method: speculative BASIC/server assumptions were replaced by inspected NABU/z88dk behavior, a verified transport foundation, a bounded Gateway protocol, native C/assembly, repeatable builds, MAME qualification, physical-hardware acceptance, and Git release discipline.

Figure 2. Project milestone sequence from the original forum proposal through the local v1.0 freeze and the next formal publication step.

2. What Version 1.0 Actually Is

Command Center is a two-part system. The modern Gateway performs HTTPS/API access, provider-specific parsing, geospatial reduction, caching, data normalization and format conversion. The NABU client consumes only bounded records through the established RetroNET Store path and renders the period-inspired user interface. Program loading remains a separate Internet Adapter function; it is not treated as the application data transport.

Figure 3. Production architecture. The Gateway absorbs modern service complexity while the NABU remains a deterministic native client.

2.1 Production modules

Subsystem	v1.0 status	Production behavior
Atomic Clock	LIVE / VERIFIED	Network time service presented as a persistent global service.
Local Weather	LIVE / VERIFIED	Current conditions, wind, pressure, trends, ZIP/location selection, and truthful source/state reporting.
Weather Alerts	LIVE / VERIFIED	Bounded NWS alert presentation integrated with Weather.
Earthquake	LIVE / VERIFIED	USGS-backed local/global views with magnitude, depth, age and target selection.
Space Weather	LIVE / VERIFIED	NOAA/SWPC values, severity and bounded historical trends.
Satellite / ISS	LIVE / VERIFIED	ISS/NORAD 25544 live position/state with vector orbital presentation.
Airspace / ADS-B	LIVE / VERIFIED	ADSB.lol-backed bounded aircraft state in the owner-approved 3D perspective/vector renderer.
NABU Task Manager	LIVE / VERIFIED	Three internal pages using truthful LIVE/CALC/MAP/CONST/UNAVAILABLE diagnostics.
Music / Sound	LIVE / VERIFIED	AY UI cues plus bounded Music playback; B toggles Music and keyboard C remains unbound.
Startup / Help / Safe Idle	VERIFIED	v1.0 splash, any-key bypass, compact production Help and safe return behavior.

2.2 Provider and data provenance at a glance

Service	Production source	Role
Time	NIST Internet Time Service	Gateway-side acquisition; bounded time record.
Weather / locality	National Weather Service + Census ZCTA support	Current observations and friendly ZIP/location resolution.
Weather alerts	National Weather Service	Official alert feed reduced for NABU presentation.
Earthquake	USGS	GeoJSON data reduced to local/global bounded event state.
Space weather	NOAA / SWPC	Current indices, solar-wind/field state and bounded history.

Service	Production source	Role
Satellite / ISS	WhereTheISS	Live ISS state for NORAD 25544.
Airspace / ADS-B	ADSB.lol	Regional point query around selected location; bounded aircraft set.
Music	Mutopia source material + Mido 1.3.3	Gateway conversion to compact MU01 event stream.

3. Production Interface and Hardware Evidence

The following captures show the actual production interface using MAME development/acceptance captures and Gateway evidence rather than conceptual mockups. Physical-NABU photographs have been removed from this public-facing edition for privacy; the recorded physical-hardware verification status remains unchanged.

Figure 4. MAME development/acceptance dashboard capture showing the owner-frozen Page 1 visual language and module layout.	Figure 5. MAME Local Weather / Trends screen showing live-state presentation, bounded graph region and source/status fields.
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Figure 6. MAME Earthquake Local Terrain view with bounded 3D/vector presentation and target details.
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3.1 Internal diagnostics remain truthful

4. Version 1.0 Freeze and Reproducibility Record

The release candidate is no longer a floating development snapshot. Codex completed the Owner-authorized local freeze after first verifying the exact accepted bytes and source identity. No rebuild was performed during the freeze, so the hardware-accepted binary remained unchanged.

Field	Frozen v1.0 evidence
BUILD_ID	NCC-SPLASH-260826-LV2
000001.bin	59,501 bytes
000001.NABU	59,501 bytes
Artifact relationship	Byte-identical - VERIFIED
SHA-256	5D27DFBB3ED8DCD96AF828CECC7550EF546B07C10509E20A71FECFCF0CC89087
BSS_END / initial SP	\$FC7A / \$FF00
Static-to-stack address-space margin	646 bytes
MAME	OWNER VERIFIED
Physical NABU	OWNER VERIFIED
Integrated runtime	OWNER VERIFIED
Golden Transport endurance	5,000 / 5,000 bounded reads
Separate final LV2 endurance duration	Not claimed

4.1 Git freeze state

Item	Value	Meaning
Accepted-state commit	6d543f480e247a8cc7d5b7996b9f4a0235c8f5a2	Captures owner-accepted LV2 source, Gateway integration, tests, artifacts and build evidence.
Release organization commit	2566641c99b7b725a4f08c6c7ace9c7ce1bb2496	Adds release packaging, indexes, manifests and unchanged final documents.
Release branch	release/v1.0	Points to the final organized freeze commit.
Annotated tag	ncc-command-center-v1.0-physical-verified	Tags the final organized commit; annotation records LV2 hash and Owner MAME/physical acceptance.
Tag object	e4d79f483ffdd37db2a0cb14b76a9f1b8daaf4	Local annotated tag object.
Tracked working tree	CLEAN	Historical/unrelated owner material was intentionally left untracked and untouched.
Git integrity	PASS	git diff --check and git fsck --full both passed.
Remote state	None configured	No push was performed.

4.2 Release directory

The repository now contains an authoritative releases/v1.0 area with README, release manifest, version and build provenance records, acceptance status, known limitations, SHA256SUMS, documentation index, exact accepted .bin/.NABU artifacts, and retained map/listing/symbol evidence. The three final DOCX publications were copied unchanged and verified byte-identical to their supplied originals.

4.3 Gateway behavior is visible and auditable

Figure 7. Integrated Gateway console during development/acceptance, showing bounded publication activity across Command Center services. The Gateway is the modern-data service owner; the NABU remains the native thin client.

4.4 Licensing and redistribution

MIT LICENSE — Derek-owned NABU Command Center source and documentation are released under the MIT License, Copyright (c) 2026 Derek Leger. The license permits use, copying, modification, merging, publication, distribution, sublicensing, and sale subject to preservation of the copyright and permission notice. Third-party components, tools, data services, and referenced material remain subject to their own licenses or terms and are not relicensed by this notice.

5. How Version 1.0 Will Be Formally Published

The remaining release work is deliberately short and controlled. I do not intend to reopen the v1.0 implementation simply because the repository is becoming public. The physical-verified tag and accepted native binary are evidence anchors; the publication pass is about licensing, packaging, discoverability and an end-to-end public-user check.

Figure 8. Formal publication sequence. Each step preserves the frozen native artifact and adds only the distribution evidence needed for a trustworthy community release.

Stage	Method	Release control
1. Licensing / provenance	COMPLETE. Derek-owned project material is released under the MIT License, Copyright (c) 2026 Derek Leger. Third-party boundaries and attribution remain governed by their own terms and notices.	MIT LICENSE SELECTED / THIRD-PARTY BOUNDARIES PRESERVED.

Stage	Method	Release control
2. Public Git repository	Create the reviewed public repository only after Owner authorization. Publish the release-ready source/Gateway/docs structure without sweeping in private caches, scratch material or unrelated evidence.	The local repository has no remote and no push has occurred.
3. GitHub v1.0 release	Create the public release from the frozen release state/tag relationship and attach the exact .NABU artifact, release ZIP and checksum/manifest material. State BUILD_ID and SHA-256 explicitly.	The public binary must match the hardware-accepted 59,501-byte artifact.
4. NABU Cloud staging	Use the verified contributor workflow, create the NABU Channel under My NABU Programs, upload the exact .nabu binary, add description/screenshots, create the four-tile menu icon, select an existing appropriate group, and keep the entry unpublished during QA.	The Cloud listing distributes the native program; the companion Gateway remains in the public release package.
5. Publish and verify	Turn the NABU Cloud entry to Published, refresh/restart the Internet Adapter through the documented current procedure, confirm the entry appears in the public catalog, boot it by the normal public path, and confirm visible v1.0 identity. Separately test the public Gateway package from a clean folder using only published instructions.	Only after this end-to-end check is PUBLIC NABU DISTRIBUTION VERIFIED appropriate.
6. Community announcement	Return to the original forum thread and publish the completed-project update with the NABU Cloud and source/release links, Gateway requirement, final hash, MAME/physical verification statement and known limitations.	This closes the loop from the December 2023 proposal to the released native application.

PUBLICATION MODEL The NABU Cloud entry is the community-facing distribution point for the native .NABU program. The public Git release is the reproducibility and companion-Gateway home. Both must identify the same v1.0 release and clearly explain that live services require the matching Gateway.

5.1 What a community user should expect

- Select NABU Command Center from the community catalog through the NABU Internet Adapter.
- Obtain the matching v1.0 Gateway package and documentation from the public release.
- Configure/start the Gateway using the published example configuration and current documented Store workflow.
- Boot the native channel, confirm the Version 1.0 splash/build identity, enter the desired ZIP/location, and allow the live records to populate.
- Use the published manual for module controls, refresh behavior, status meanings, Music, Task Manager, Help and troubleshooting.

5.2 Current official/community publication references

Reference	Location	Use
NABU Cloud - About This Website	https://nabu.ca/about-this-website	Contributor verification, My NABU Programs, .nabu upload, screenshots/icon, groups, publish toggle and catalog behavior. Checked in the release plan on 26 Aug 2026.
NABU Channels / software portal	https://cloud.nabu.ca/software	Current native channel/catalog context for the NABU Internet Adapter.
NABU Downloads	https://nabu.ca/downloads	Current project download/developer resources.
GitHub Releases documentation	https://docs.github.com/en/repositories/releasing-projects-on-github/about-releases	Tag-based public release model and attached binary assets.
Original project forum thread	https://forums.nabu.ca/viewtopic.php?t=206	December 23, 2023 proposal that will receive the completion update.

6. Near-Term Publication Expectations

At this point I consider Version 1.0 engineering complete at the frozen local-release level. The next pass should not be measured in new modules; it should be measured in publication gates. The licensing/provenance gate is now closed under the MIT License for Derek-owned material. The remaining publication sequence is the public repository/release, NABU Cloud staging, end-to-end public-path verification and forum announcement can proceed in sequence.

Gate	State	Expectation
Native v1.0 implementation	COMPLETE / FROZEN	Do not reopen without a real release-blocking defect.
MAME acceptance	OWNER VERIFIED	Exact LV2 line accepted.
Physical NABU acceptance	OWNER VERIFIED	Exact accepted binary lineage verified on real hardware.
Git organization / local freeze	VERIFIED	release/v1.0 and annotated physical-verified tag exist; integrity checks passed.
Documentation package	BUILT	Master Production Engineering White Paper v1.1, User Manual & Operator Guide, and updated roadmap included.
License / third-party notices	COMPLETE — MIT	Derek-owned material: MIT License, Copyright (c) 2026 Derek Leger; third-party terms preserved.
Public Git / GitHub release	PENDING	No remote configured; no push performed.
NABU Cloud publication	PENDING	Stage unpublished first, then publish after metadata/hash review.
Public-path acceptance	PENDING	Verify catalog discovery, launch and clean Gateway installation.
Forum/community announcement	PENDING	Publish only after public links and public-path verification are real.

7. Closing Message to the NABU Community

When I first posted the Weather and News idea in 2023, I was trying to answer a simple question: can the NABU be useful as a live information terminal again, not just as a preserved endpoint for old software? The answer is yes, but

not by pretending the Z80 is a modern web client.

Command Center works because the architecture respects the machine. A modern Gateway handles volatile Internet services; the NABU receives compact records and presents them through native software built around its real constraints. The result is a living information console that still feels like NABU software.

I also wanted the release to be reproducible. The final native binary is hashed and frozen, the source and Gateway state are in Git, the release branch and physical-verified tag are recorded, the build/map evidence is retained, the user and engineering documentation are complete, and the final public package will carry the same identity. If somebody else builds it, studies it, or reports a defect, we can talk about the same version rather than an ambiguous moving target.

The formal community release is next. The license/provenance gate is now closed under MIT. The publication sequence is to publish the reviewed source and companion Gateway, stage the exact .NABU through NABU Cloud, verify the public distribution path as an ordinary user would, and then return to the original forum thread with the release links, checksum and operating guidance. Version 1.0 will then become immutable; any later fix or provider adaptation will be versioned rather than silently replacing the original release.

That is the standard I want attached to this project: native NABU software, real data, bounded interfaces, evidence-backed claims, reproducible artifacts, and a release that another NABU owner can actually understand and run.

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NABU Command Center v1.0

Appendix A. Frozen Release Snapshot

Field	Value
Product	NABU Command Center
Version	1.0
Build	NCC-SPLASH-260826-LV2
Artifact	000001.NABU - 59,501 bytes
SHA-256	5D27DFBB3ED8DCD96AF828CECC7550EF546B07C10509E20A71FECFCF0CC89087
Release branch	release/v1.0
Final organized commit	2566641c99b7b725a4f08c6c7ace9c7ce1bb2496
Annotated tag	ncc-command-center-v1.0-physical-verified
MAME	OWNER VERIFIED
Physical NABU	OWNER VERIFIED
Public remote	Not yet configured
License	MIT License — Copyright (c) 2026 Derek Leger; third-party material retains its own terms.
Public distribution	PENDING formal publication sequence

Appendix B. Documentation Included in the Freeze

Document	Purpose
NABU_Command_Center_Master_Production_Engineering_White_Paper_v1.1.docx	Complete production engineering record and reconstruction reference.
NABU_Command_Center_v1.0_User_Manual_and_Operator_Guide.docx	37-page operator/user guide with real development and physical-acceptance imagery.
NABU_COMMAND_CENTER_Engineering_White_Paper_Roadmap_2026-08-26_Updated.docx	Preserved roadmap plus 26 Aug as-built overlays.
NABU_Command_Center_v1.0_Final_Production_Release_and_Publication_Plan.docx	Publication procedure, release/legal gates, NABU Cloud process and post-publication verification plan.

Source basis: Owner/Codex v1.0 local freeze report (27 Aug 2026); frozen project documentation listed above; current publication plan web-source checks dated 26 Aug 2026; user-supplied forum and development/acceptance captures. Historical screenshots are labeled as such and are not used to override final artifact identity.